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Human Kinetics
ORIGINAL INVESTIGATION

Maintenance of Velocity and Power With Cluster Sets During High-Volume Back Squats

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Purpose: To compare the effects of a traditional set structure and 2 cluster set structures on force, velocity, and power during back squats in strength-trained men. ***Methods:*** Twelve men (25.8 ± 5.1 y, 1.74 ± 0.07 m, 79.3 ± 8.2 kg) performed 3 sets of 12 repetitions at 60% of 1-repetition maximum using 3 different set structures: traditional sets (TS), cluster sets of 4 (CS4), and cluster sets of 2 (CS2). ***Results:*** When averaged across all repetitions, peak velocity (PV), mean velocity (MV), peak power (PP), and mean power (MP) were greater in CS2 and CS4 than in TS ($P < .01$), with CS2 also resulting in greater values than CS4 ($P < .02$). When examining individual sets within each set structure, PV, MV, PP, and MP decreased during the course of TS (effect sizes 0.28–0.99), whereas no decreases were noted during CS2 (effect sizes 0.00–0.13) or CS4 (effect sizes 0.00–0.29). ***Conclusions:*** These results demonstrate that CS structures maintain velocity and power, whereas TS structures do not. Furthermore, increasing the frequency of intraset rest intervals in CS structures maximizes this effect and should be used if maximal velocity is to be maintained during training.

Keywords: intraset, rest, hypertrophy, traditional sets, fatigue

To increase mechanical power, athletes perform resistance-training exercises with maximum intended concentric velocities.[1–4] Traditionally, repetitions in each set are performed in sequence with no rest between repetitions: A structure that has been defined as a traditional set structure.[5–7] When using traditional sets, concentric velocity decreases as the number of repetitions increases.[7–11] Because the external mechanical power of a lift is the product of force and velocity, a reduction in velocity results in a concomitant decrease in power if force remains the same. Patterns of reduced velocity and power across traditional sets have been documented in a variety of complex lifting movements such as the jump squat,[2,12] leg press,[8,13] and weightlifting exercises.[7,10] These decreases are magnified when multiple sets contain a high number of repetitions.[14]

In high-volume resistance training, there are large reductions in movement velocity and power output within sets and across multiple sets.[9,14] As evidence, Sánchez-Medina et al[14] reported a 46% reduction in mean velocity across 3 sets of 12 back squat repetitions performed to failure with 5 minutes of interset rest. Ultimately, traditional set structures that contain many repetitions result in accumulated fatigue, which may result in a reduced capacity to maintain a high velocity throughout a training session.

One potential explanation for the decrease in movement velocity, and ultimately power output, is the combination of reduced adenosine triphosphate (ATP) and phosphocreatine (PCr) availability[8,9,13] and increased lactate and ammonia accumulation[8] that occurs when many repetitions are performed over multiple sets. However, if the number of repetitions per set is reduced and more frequent rest periods are provided (eg, 5 sets of 10 vs 10 sets of 5), the phosphagen system appears to be able to meet the energetic demands of the exercise.[8,9,13] Specifically, a maintenance of ATP and PCr availability enables movement velocity to remain near maximal. From a programmatic perspective, placing short rest intervals within traditional sets to create what is commonly referred to as a cluster set[5–7] may assist in maintaining movement velocity across sets and an entire exercise session.

Since the original investigation of cluster set structures by Haff et al,[7] research has determined that various intraset and inter-repetition rest periods acutely maintain force, velocity, and power, blunting the decline that is normally seen when using traditional sets.[2,11,12,15] For example, Hardee et al[10] reported that by adding 20-second interrepetition rest intervals, movement velocity and power output decreased less over 3 sets of 6 repetitions during the power clean when compared with 6 successive repetitions. When interrepetition rest was extended to 40 seconds, the peak velocity and power output of each repetition was better maintained than when 20 seconds of interrepetition rest was used.[10] Collectively, the acute cluster set literature uses set structures with low repetition ranges that are typically associated with maximal strength or power training, leaving the acute effects of cluster sets on high-volume resistance training relatively unknown.

Previous cluster set literature has examined peak velocity (PV),[2,7,10,12,15] mean velocity (MV),[16,17] peak power (PP),[2,10,12,15,18] mean power (MP),[16,19–23] peak force (PF),[2,10,12,15] and mean force (MF),[16,19,22] but none of these studies investigated all of these variables within the same study. In addition, due to the limited number of studies examining the effect of high-volume cluster set structures on these variables across multiple sets, there is a need for research on this topic. Therefore, the primary purpose of this investigation was to compare the effects of traditional sets and 2 different cluster set structures on PF, MF, PV, MV, PP, and MP during a high-volume back squat session performed at maximal velocity in strength-trained men. We hypothesized that PV, MV, PP, and MP would be different between the 3 protocols, but that PF and MF would not.

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Methods

Subjects

Twelve strength-trained men participated in this study (age 25.8 ± 5.1 y, height 1.74 ± 0.07 m, body mass 79.3 ± 8.2 kg), had at least 6 months of strength training experience using the full back squat exercise, and could back squat at least 150% of their body mass. Participants were screened using medical history questionnaires and were excluded if they reported any recent musculoskeletal injuries. Participants averaged a 1-repetition maximum (1RM) of 148.8 ± 11.5 kg, a 1RM-to-body-mass ratio of 1.90 ± 0.23, and a peak knee flexion angle at the bottom of the squat of 122.9° ± 11.3°. All procedures were carried out in accordance with the Declaration of Helsinki and were approved by the university human research ethics committee. All participants gave written informed consent before participation.

Design

Testing occurred over 4 sessions: a 1RM session and 3 experimental sessions. Using a randomized design, participants completed each of the 3 protocols on separate days, 48 to 96 hours apart, and were instructed to refrain from any type of fatiguing lower body activity for the duration of the study. Each protocol consisted of 3 sets of 12 back squats using 60% 1RM, and each of the protocols consisted of different set structures defined by different rest periods (Figure 1). Previous studies have shown that a load of 70% 1RM equates to roughly a 12RM in the Smith machine back squat[14] and that 70% 1RM cannot be maintained during 4 sets of 10 barbell back squats with 2 minutes of interset rest.[16] During pilot testing, participants were only able to complete all 36 repetitions with 60% 1RM during the traditional set protocol (TS), described in detail later. Therefore, to ensure that training to failure was avoided, an external load of 60% was chosen. As a result, all participants successfully completed all 36 repetitions in all 3 protocols.

Methodology

Repetition-Maximum Testing: Session 1. Participant height was measured using a wall-mounted stadiometer, and body mass was measured using a calibrated electronic scale. Participants then completed a warm-up on a cycle ergometer for 5 minutes at 100 W at 60 revolutions per minute. Warm-up squats consisted of 10 bodyweight squats followed by 8, 5, and 3 repetitions at 25%, 50%, and 60% estimated 1RM, respectively. Back squat 1RM was then assessed starting at 85% estimated 1RM, and load was progressively increased until the 1RM was achieved using previously established methods.[24] Participants' heel and toe locations were recorded on the force plate using a horizontal-vertical grid intersecting every 1 cm and were maintained throughout all testing sessions.

Experimental Testing: Sessions 2, 3, and 4. Experimental testing sessions 2, 3, and 4 used the same warm-up as session 1, but included warm-up loads based off of actual 1RM. Each session consisted of a different, randomized protocol. Specifically, TS consisted of 3 sets of 12 repetitions at 60% 1RM with seated interset rest intervals of 120 seconds (Figure 1[A]). Cluster sets of 4 (CS4) used the same structure as TS with the exception of an additional 30 seconds of standing, but unloaded intraset rest after the fourth and eighth repetition of each set (Figure 1[B]). Cluster sets of 2 (CS2) used the same structure as TS with the exception of an additional 30 seconds of standing, but unloaded intraset rest after the second, fourth, sixth, eighth, and tenth repetitions of each set (Figure 1[C]).

During these sessions, an investigator provided participants with a verbal countdown descending from 10 seconds, and the participant unracked the bar when the countdown reached "zero." In an attempt to maximize back squat peak power, participants were instructed to perform the concentric phase of each squat as quickly as possible to a standing position[25] while the barbell was lowered under control. To ensure that the eccentric velocity was kept constant, each participant used a self-regulated eccentric velocity (no significant differences in eccentric velocity within participants between conditions, $P = .18$, $d = 0.29$). Participants were instructed to squat "all the way down" and "explode out of the bottom." Squat depth was monitored using real-time visual displacement curves to encourage that all repetitions were performed to approximately the same depth (no significant differences in depth within participants between conditions, $P = .98$, $d = 0.02$). During all repetitions, the feet were required to maintain contact with the force plate (eg, no jumping or lifting of the heels),[25] and a slight pause was required at the conclusion of each repetition to ensure full hip and knee

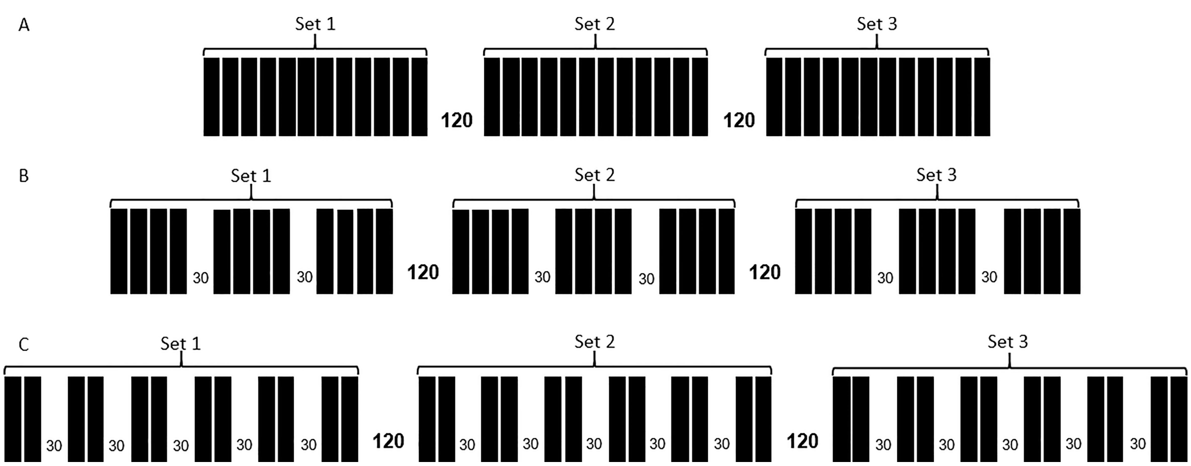

Figure 1 — Set structure protocols. (A) Traditional sets, 33 sets of 12 with 120 seconds of interset rest. (B) Cluster sets 4, 3 sets of 12 with 120 seconds of interset rest and 30 seconds of intraset rest after the fourth and eighth repetition of each set. (C) Cluster sets 2, 3 sets of 12 with 120 seconds of interset rest and 30 seconds of intraset rest after the even-numbered repetitions of each set.

extension. Sessions 3 and 4 were performed exactly the same as session 2, with the exception of completing the remaining 2 testing protocols in random order.

Force, Velocity, and Power Data Acquisition. All kinematic and kinetic data were collected using methodology similar to previous research.[26] Briefly, all squats were performed on a force plate (AMTI BP12001200, Advanced Mechanical Technologies Inc, Watertown, MA, USA) to obtain PF and MF, and 2 linear position transducers (LPTs) were attached to each side of the barbell originating from the top of the squat rack (PT5A-250, Celesco Transducer Products, Chatsworth, CA, USA) to obtain PV and MV. All force plate and LPT data were collected via a BNC-2090 interface box with an analog-to-digital card (NI-6014; National Instruments, Austin TX, USA) and sampled at 1000 Hz. A customized LabVIEW program, Version 14.0 (National Instruments) was used to collect and analyze all force and displacement data. Signals were filtered using a fourth order-low pass Butterworth filter with a cutoff frequency of 50 Hz. External mechanical MP and PP of the system were calculated by direct measurement of ground reaction force and bar velocity. The retraction tension of the 4 LPTs was 23.0 N, which was accounted for in all calculations, and all variables were assessed during the concentric phase. The effect of set structure on MV, PV, MP, and PP across each protocol was determined by a percent decline from the first to 36th repetition using the following equation:

$$\text{Percent decline} = [(\text{repetition}_{36} - \text{repetition}_1)/\text{repetition}_1] \times 100$$

Furthermore, the ability to maintain MV, PV, MP, and PP during all repetitions within each set was assessed using the following equation:

$$\text{Maintenance}_{\text{set}} = 100 - [(\text{mean}_{\text{set}} - \text{repetition}_1)/\text{repetition}_1] \times 100$$

As a result, the variables of MV and PV decline (MVD and PVD, respectively), MP and PP percent decline (MPD and PPD, respectively), MV and PV maintenance (MVM and PVM, respectively), and MP and PP maintenance (MPM and PPM, respectively) were calculated.

Statistical Analyses

Means and SDs were calculated for all variables. Individual 3 × 3 (protocol × set) repeated-measures analysis of variance (ANOVA) was used to compare means for all variables except percent decline variables and protocol time. In the event of significant main effects and interactions, a Holm's Sequential Bonferroni follow-up test was performed to control for type I error and assess pairwise comparisons. For decline variables and protocol time, paired *t* tests with a Holm's Sequential Bonferroni follow-up were used to control for type I error. Effect sizes were calculated using Cohen's d and can be interpreted as small, $d = 0.2$; medium, $d = 0.5$; and large, $d = 0.8$. Significance was set at $P \leq .05$ for all tests. All statistical analyses were performed using SPSS, version 22.0 (IBM, Armonk, NY).

Results

Mean ± SD for MF, PF, MV, PV, MP, and PP are presented in Table 1. Repetition PV and PP are presented in Figure 2; MV and MP are in Figure 3. Significant interactions are described in the text, whereas particular P values and effect sizes are shown in Table 2. There were no main effects for protocol when examining MF ($P = .884$, $d < 0.01$) and PF ($P = .264$, $d < 0.01$) (Table 1). As intended by design, the duration of TS (6:02 ± 0:16 min) was less than CS4 (9:34 ± 0:21 min, $P < .001$, $d = 17.78$) and CS2 (15:12 ± 0:28 min, $P < .001$, $d = 39.91$), and CS4 was less than CS2 ($P < .001$, $d = 23.27$).

There was a protocol × set interaction for MV ($P = .001$, $d = 0.29$) and PV ($P < .001$, $d = 0.20$) (Table 3). There was a protocol × set interaction for MVM ($P = .001$, $d = 0.36$) and PVM ($P < .001$, $d = 0.45$) (Figures 4 and 5). For MVD, there was a difference between CS2 and TS ($P = .002$, $d = 1.66$), CS2 and CS4 ($P = .034$, $d = 0.48$), and CS4 and TS ($P = .002$, $d = 1.38$) (Figure 6). For PVD, there was a difference between CS2 and TS ($P = .004$, $d = 1.47$) and CS4 and TS ($P = .003$, $d = 1.29$), but not between CS2 and CS4 ($P = .184$, $d = 0.41$) (Figure 6).

There was a protocol × set interaction for MP ($P = .003$, $d = 0.25$) and PP ($P < .001$, $d = 0.25$) (Table 3). There was a protocol × set interaction for MPM ($P = .002$, $d = 0.33$) and PPM ($P < .001$, $d = 0.42$) (Figures 4 and 5). For MPD, there was a difference between CS2 and TS ($P = .003$, $d = 1.67$), CS2 and CS4 ($P = .014$, $d = 0.56$), and CS4 and TS ($P = .002$, $d = 1.36$) (Figure 6). For PPD, there was a difference between CS2 and TS ($P = .003$; $d = 1.49$) and CS4 and TS ($P = .002$, $d = 1.40$), but not between CS2 and CS4 ($P = .269$, $d = 0.22$) (Figure 6).

Discussion

The major findings of the current study were that the inclusion of 30-second intraset rest intervals during cluster sets alleviated fatigue-induced decreases in velocity and power over 3 sets. Specifically, more frequent intraset rest during CS2 resulted in greater MV, PV, MP, and PP than CS4 while also resulting in less MVD and MPD. As such, velocity and power were maximized when intraset rest was most frequent and the total rest time was greatest.

Table 1 Force, Velocity, and Power for All 36 Repetitions Within Each Protocol, Mean ± SD for All 36 Repetitions

	Cluster sets 2	Cluster sets 4	Traditional sets
Mean concentric force (N)	1603.6 ± 136.1	1601.4 ± 146.9	1606.1 ± 142.6
Peak concentric force (N)	2526.2 ± 237.7	2519.9 ± 248.7	2557.8 ± 248.2
Mean concentric velocity (m/s)	0.80 ± 0.08*†	0.77 ± 0.06*	0.72 ± 0.05
Peak concentric velocity (m/s)	1.37 ± 0.18*‡	1.30 ± 0.14*	1.21 ± 0.13
Mean concentric power (W)	1261.2 ± 115.7*†	1213.07 ± 92.3*	1143.8 ± 80.4
Peak concentric power (W)	2563.0 ± 403.4*‡	2412.90 ± 306.6*	2207.7 ± 243.0

*Significantly greater than traditional set ($P < .01$); significantly greater than cluster set 4, †$P < .01$, ‡$P < .05$.

Table 2 Protocol × Set and Set × Protocol Comparisons, *P* (Effect Size, *d*)

		Protocol Comparisons			Set Comparisons				
		CS2–CS4	CS4–TS	CS2–TS	1–2	2–3	1–3		
MV	set 1	.040 (0.66)	.014 (0.40)	.003 (0.99)	.876 (0.13)	.199 (0.00)	.544 (0.13)	CS2	MV
	set 2	.008 (0.57)	.001 (0.54)	<.001 (1.05)	.127 (0.18)	.490 (0.00)	.131 (0.16)	CS4	
	set 3	.004 (0.53)	<.001 (1.00)	<.001 (1.46)	.069 (0.40)	.012 (0.66)	.010 (0.99)	TS	
PV	set 1	.041 (0.45)	.048 (0.39)	.004 (0.76)	.892 (0.00)	.549 (0.00)	.824 (0.00)	CS2	PV
	set 2	.030 (0.42)	.003 (0.62)	.001 (0.99)	.969 (0.00)	.544 (0.00)	.678 (0.00)	CS4	
	set 3	.006 (0.41)	<.001 (1.00)	<.001 (1.36)	.019 (0.30)	.017 (0.43)	.003 (0.74)	TS	
MP	set 1	.034 (0.37)	.006 (0.56)	.001 (0.87)	.773 (0.03)	.141 (0.08)	.392 (0.11)	CS2	MP
	set 2	.005 (0.49)	.001 (0.63)	<.001 (1.06)	.064 (0.21)	.334 (0.09)	.046 (0.29)	CS4	
	set 3	.005 (0.49)	<.001 (1.02)	<.001 (1.41)	.085 (0.33)	.011 (0.62)	.011 (0.89)	TS	
PP	set 1	.046 (0.37)	.060 (0.45)	.009 (0.72)	.582 (0.06)	.741 (0.05)	.588 (0.07)	CS2	PP
	set 2	.025 (0.44)	.005 (0.67)	.001 (1.03)	.834 (0.02)	.996 (0.00)	.888 (0.02)	CS4	
	set 3	.016 (0.42)	<.001 (1.04)	<.001 (1.40)	.060 (0.28)	.014 (0.52)	.004 (0.79)	TS	
MVM	set 1	.400 (0.23)	.005 (1.15)	.053 (0.80)	.911 (0.02)	.244 (0.09)	.600 (0.11)	CS2	MVM
	set 2	.989 (0.00)	.012 (0.97)	.006 (0.98)	.134 (0.24)	.440 (0.10)	.130 (0.35)	CS4	
	set 3	.940 (0.01)	.006 (1.24)	0.010 (1.21)	.076 (0.41)	.009 (0.59)	.009 (0.92)	TS	
PVM	set 1	.715 (0.09)	.037 (0.70)	.064 (0.53)	.977 (0.01)	.509 (0.09)	.723 (0.10)	CS2	PVM
	set 2	.897 (0.04)	<.001 (0.99)	.004 (0.92)	.887 (0.03)	.491 (0.10)	.581 (0.15)	CS4	
	set 3	.581 (0.15)	.001 (1.32)	.002 (1.37)	.023 (0.50)	.015 (0.57)	.003 (1.03)	TS	
MPM	set 1	.442 (0.22)	.005 (1.26)	.025 (0.92)	.742 (0.06)	.174 (0.11)	.401 (0.18)	CS2	MPM
	set 2	.867 (0.04)	.016 (0.99)	.006 (1.00)	.061 (0.31)	.320 (0.13)	.048 (0.48)	CS4	
	set 3	.831 (0.04)	.009 (1.64)	.011 (1.55)	.086 (0.37)	.008 (0.71)	.009 (1.17)	TS	
PPM	set 1	.622 (0.09)	.027 (0.66)	.070 (0.52)	.535 (0.15)	.718 (0.03)	.542 (0.17)	CS2	PPM
	set 2	.701 (0.11)	.001 (0.86)	.002 (0.92)	.794 (0.05)	.900 (0.02)	.782 (0.06)	CS4	
	set 3	.532 (0.15)	.001 (1.30)	.001 (1.37)	.074 (0.37)	.011 (0.58)	.004 (0.95)	TS	

Abbreviations: CS2, cluster sets 2; CS4, cluster sets 4; TS, traditional sets; MV, mean velocity; PV, peak velocity; MP, mean power; PP, peak power; MVM, mean velocity maintenance; PVM, peak velocity maintenance; MPM, mean power maintenance; PPM, peak power maintenance.

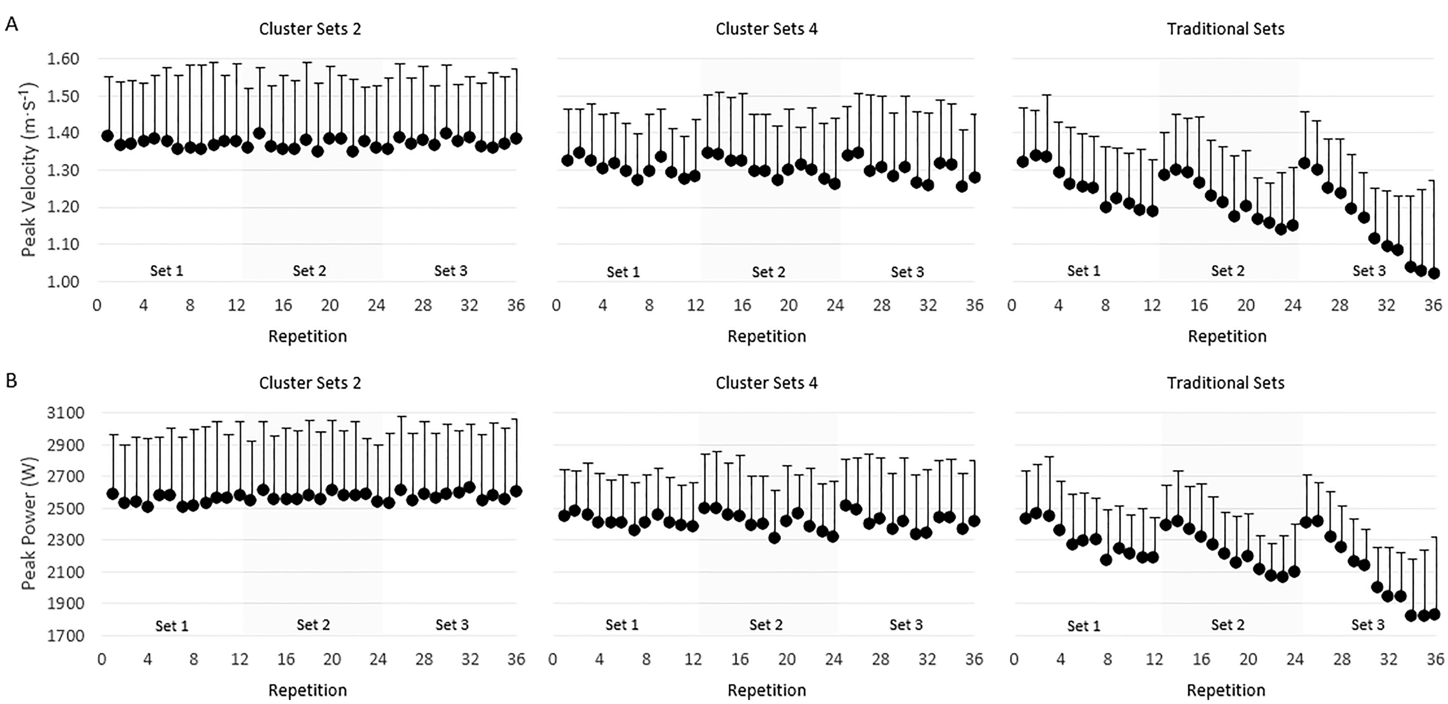

Figure 2 — (A) Peak velocity and (B) peak power during 3 sets of 12 repetitions; 36 total repetitions. Peak force is not shown, as it did not significantly change across sets or repetitions.

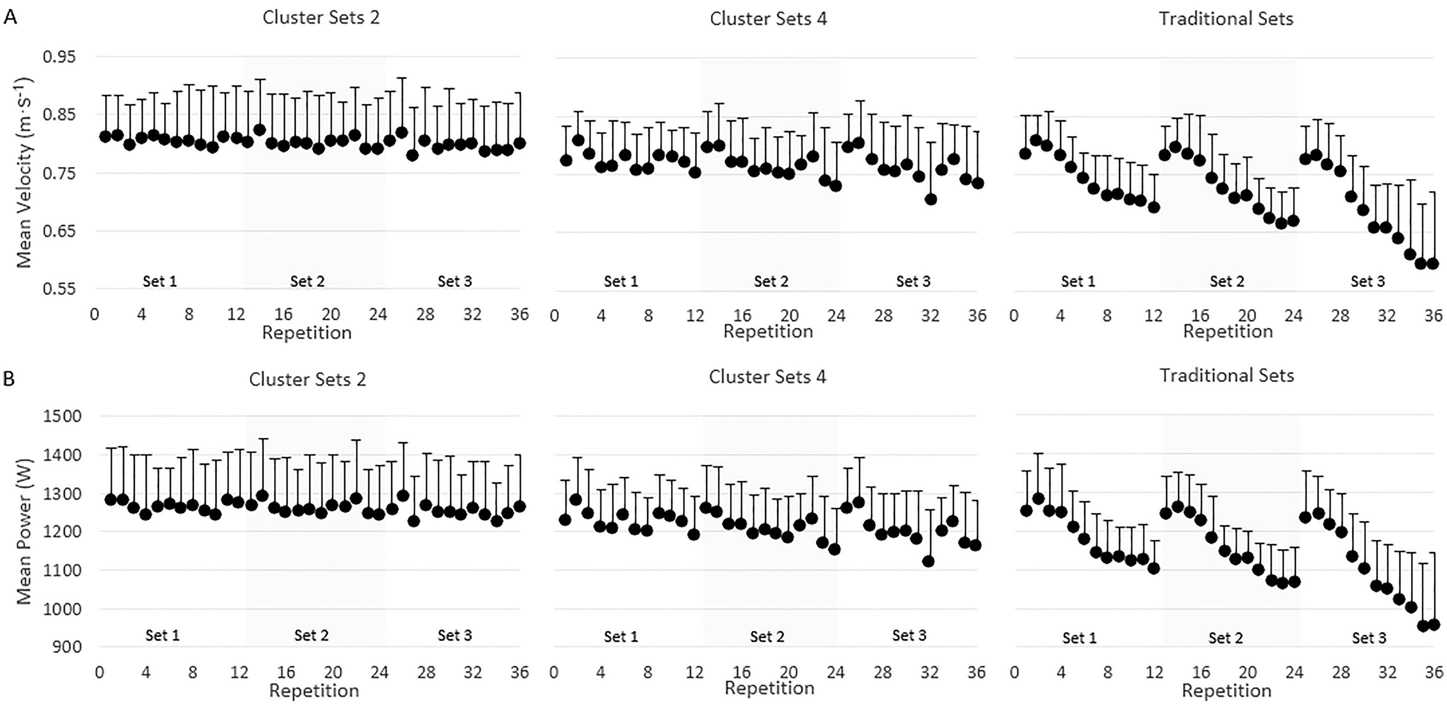

Figure 3 — (A) Mean velocity and (B) mean power during 3 sets of 12 repetitions; 36 total repetitions. Mean force is not shown as it did not significantly change across sets or repetitions.

Table 3 Force, Velocity, and Power of All 12 Repetitions Within Each Set During Each Protocol, Mean ± SD

		Cluster sets 2	Cluster sets 4	Traditional sets
Mean concentric velocity, m/s	set 1	0.81 ± 0.07*†	0.77 ± 0.05*	0.75 ± 0.05~
	set 2	0.80 ± 0.08*†	0.76 ± 0.06*	0.73 ± 0.05~
	set 3	0.80 ± 0.08*†	0.76 ± 0.07*	0.69 ± 0.07
Peak concentric velocity, m/s	set 1	1.37 ± 0.18*	1.30 ± 0.13*	1.25 ± 0.13~^
	set 2	1.37 ± 0.18*†	1.30 ± 0.15*	1.21 ± 0.14~
	set 3	1.37 ± 0.18*†	1.30 ± 0.16*	1.15 ± 0.14
Mean concentric power, W	set 1	1266.63 ± 112.87*†	1228.61 ± 89.25*	1180.89 ± 82.61~
	set 2	1263.34 ± 123.83*†	1209.46 ± 95.80*	1154.38 ± 77.43~
	set 3	1253.68 ± 116.56*†	1201.14 ± 97.51*	1096.20 ± 107.04
Peak concentric power, W	set 1	2545.32 ± 407.92*	2416.43 ± 271.70	2296.55 ± 266.08~
	set 2	2568.79 ± 405.27*†	2411.06 ± 308.00*	2222.76 ± 252.13~
	set 3	2574.79 ± 418.41*†	2411.20 ± 351.30*	2088.80 ± 259.03

Note: Protocol differences within set were significantly greater than traditional sets.

*($P < .05$); significantly greater than cluster set 4 †($P < .05$). Set differences within protocol: ^significantly greater than set 2 ($P < .05$); ~significantly greater than set 3 ($P < .05$).

The addition of intraset rest intervals did not have an effect on force production. Specifically, neither MF nor PF changed throughout all 3 sets during any of the 3 protocols. These results are in line with other studies comparing MF during the bench press with a 6RM load[19] and PF during bodyweight[11] and 40-kg jump squats[1] using traditional and cluster set structures, but are in disagreement with others.[16] In this study,[16] the authors report that MF was greater when using cluster sets compared with traditional sets during 4 sets of 10 back squats at 70% 1RM; however, that statement is somewhat misleading because the load was decreased in the TS by an average of 8% compared with a constant load used during the cluster set structure. With a lighter load, the traditional set structure resulted in a lower MF, which should not be considered a result of a different set structure, but should be attributed to the different loads used in each protocol. Further examination of their data shows that the MF was the same between the cluster and traditional set structures before the load was decreased during the traditional sets, which agrees with the consistent MF data across all set structures in the current study.

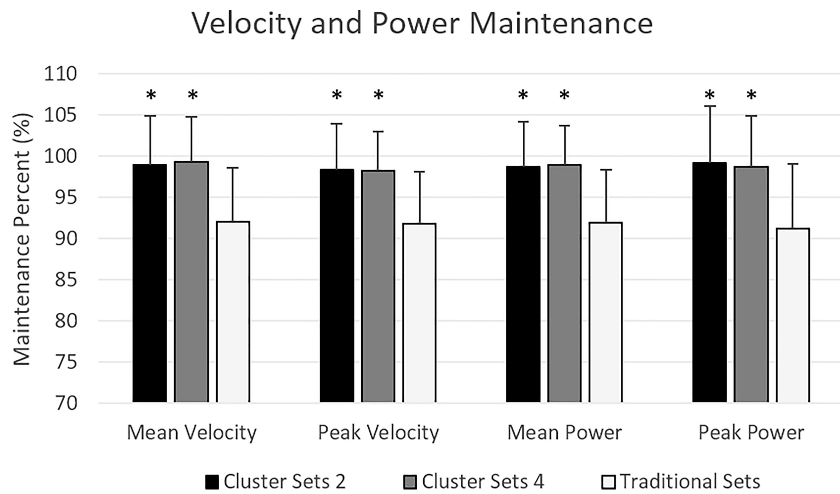

Figure 4 — Maintenance of mean velocity, peak velocity, mean power, and peak power across all 36 repetitions, expressed as a percentage of the first repetition during cluster sets 2, cluster sets 4, and traditional sets. *Significantly greater than traditional sets ($P < .01$).

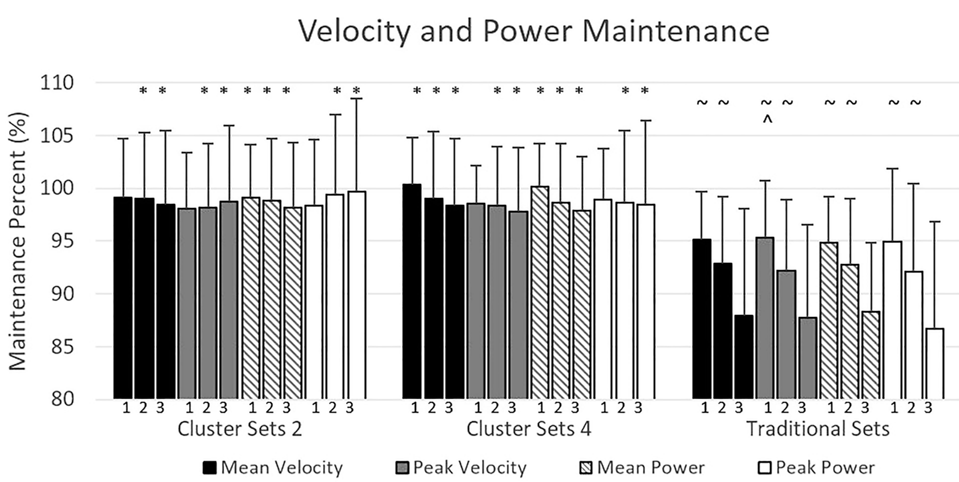

Figure 5 — Maintenance of mean velocity, peak velocity, mean power, and peak power for all 3 sets, expressed as a percentage of the first repetition during cluster sets 2, cluster sets 4, and traditional sets. Significantly greater than *traditional sets ($P < .05$), ^set 2 ($P < .05$), and ~set 3 ($P < .05$).

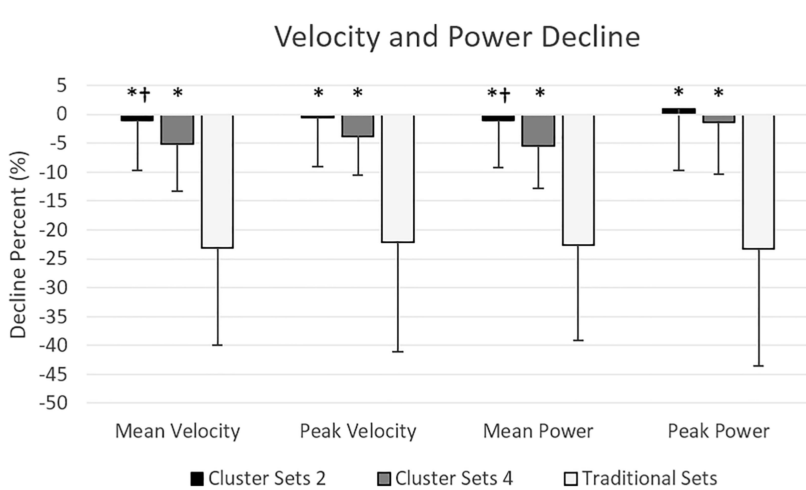

Figure 6 — Decline of mean velocity, peak velocity, mean power, and peak power, expressed as a percentage of the quotient of the 36th repetition to the first repetition during cluster sets 2, cluster sets 4, and traditional sets. *Significantly different than traditional sets ($P < .01$). †Significantly different than cluster sets 4 ($P < .05$).

Since MF and PF were not different between groups, any changes in external mechanical power output within the current study should be attributed to changes in movement velocity. Our data revealed distinct differences in velocity and power between TS, CS4, and CS2. Specifically, MV, PV, MP, and PP were less during TS compared with CS4 and CS2 and decreased during 3 sets in the TS structure, but did not decrease in CS4 or CS2. These results agree with previous studies in which velocity and power have decreased in response to traditional sets, especially in high-volume protocols.[14,16,17] Unfortunately, direct comparisons of velocity and power during cluster sets cannot be made to other studies because the only investigations using back squats during cluster set structures did not provide velocity and power data for individual sets in addition to the entire protocol.[11,16,20,23] Nonetheless, the computation of decline variables (change from the first repetition to the last) in the current study has allowed us to compare our results to those of previous research.

The TS structure in the current study resulted in an MVD, PVD, MPD, and PPD of 23% each. Similarly, Oliver et al[16] reported an MVD and PPD of ~20% across 4 sets of 10 back squats using 70% 1RM, Gorostiaga et al[8,9] showed a PPD of ~37% during 5 sets of 10 leg press to failure, and Moreno et al[12] showed that PVD was ~6% following 2 sets of 10 bodyweight jump squats. Although MVD, PVD, MPD, and PPD were 23% during TS in the current study, when cluster sets were used these values were reduced to only 1% to 5%. In agreement with our data, Hardee et al[10] showed that although PVD was 10.5% following 3 traditional sets of 6 power cleans at 80% 1RM, it was less at 5.5% and 2.5% when 20 seconds and 40 seconds of interrepetition rest was used, respectively. In addition, Hardee et al[10] also found a PPD of 18.5% during TS, but only 8.5% and 5% when 20 seconds and 40 seconds of interrepetition rest was used, respectively.

Although not measured in the current study, it can be hypothesized that intraset rest intervals may have enabled enhanced clearance of metabolic by-products and replenishment of phosphagen energy substrates in our study. A series of studies by Gorostiaga et al[8,9,13] support this by showing that when the leg press was performed to failure (5 sets of 10) decreases in power were present, which were also accompanied by increases in lactate and decreases in ATP and PCr stores. However, when the frequency of rest intervals increased (10 sets of 5), lactate levels were lower and participants maintained power, ATP stores, and PCr stores throughout the exercise session.[8,9,13] As other authors have hypothesized,[2,5,6,16,20] we believe that the addition of intraset rest in CS2 and CS4 allowed for superior replenishment of ATP and PCr, resulting in greater movement velocity during latter repetitions of each set compared with TS (Figures 2 and 3), but this idea remains unexamined in the cluster set literature.

Because many protocols required participants to train to failure,[8,9,13,16,23] and even decrease the load within the protocol to enable all repetitions to be performed, the decreases in velocity and power observed in these studies may not accurately represent what occurs during a normal resistance-training session when athletes do not train to failure and the load remains constant. Data from Sánchez-Medina et al[14] show that 3 sets of 12 squats using a 12RM load resulted in a greater decrease in mean propulsive velocity when compared with 3 sets of 10 using a 12RM load. Because our participants did not reach muscular failure, resulting in less velocity- and power-based fatigue when compared with other studies,[9,11,14] practical resistance training recommendations may be drawn from the data presented and used in situations where repeatedly training to failure may be unwarranted.[27,28]

Lastly, some studies have failed to report individual repetition data,[17,19,29] collapsed variables across sets,[2,20,30] changed the load during the protocol,[16,23] and collapsed variables between protocols,[20] making direct comparisons between studies rather difficult. Therefore, percent decline variables were reported in this discussion for the sake of keeping comparisons consistent. However, we would like to highlight the maintenance calculation, unique to the current study, as it describes what happens during the entire protocol. For example, stating that TS resulted in a 23% decline in velocity and power from the first repetition to the last would be a tenuous statement because the remaining 34 repetitions performed within the session would not be accounted for. Therefore, maintenance variables were calculated to show that participants were able to maintain velocity and power by 92% when performing TS, resulting in a decrease of only 8% when all repetitions were accounted for: much less than the 23% decline seen from the first repetition to the last. Hence, future research should address both, decline and maintenance variables, as they may each tell a different story.

Practical Applications

This study indicates that the addition of 30-second intraset rest intervals maintains velocity and external mechanical power during high volume back squats. Based upon the cluster set structures assessed, intraset rest intervals placed after every 2 repetitions is the most effective at maintaining velocity and power between sets. One drawback to the CS2 structure is that it takes more time than the CS4 and TS structures, meaning that a resistance-training program that is under strict time constraints may not be able to use CS2 structures. If a training session requires a compromise between time efficiency and fatigue management, intraset rest intervals after every fourth repetition may be used, although it is not as effective at maintaining velocity and power as intraset rest after every 2 repetitions. Future research should investigate the effects of different cluster set structures to determine the optimal number of repetitions within each cluster and the minimal amount of intraset rest needed to maintain velocity and power.

Conclusions

Based on our data, cluster sets allow strength and conditioning professionals to modify set structures to achieve different degrees of velocity-based fatigue. With this is mind, it is important for coaches to determine whether fatigue management is important and then allocate the frequency of intraset rest intervals based on the desired training outcomes.

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References

1. Behm DG, Sale DG. Intended rather than actual movement velocity determines velocity-specific training response. *J Appl Physiol*. 1993;74:359–368. PubMed
2. Hansen KT, Cronin JB, Newton MJ. The effect of cluster loading on force, velocity, and power during ballistic jump squat training. *Int J Sports Physiol Perform*. 2011;6:455–468. PubMed doi:10.1123/ijspp.6.4.455

3. González-Badillo JJ, Rodriguez-Rosell D, Sanchez-Medina L, Gorostiaga EM, Pareja-Blanco F. Maximal intended velocity training induces greater gains in bench press performance than deliberately slower half-velocity training. *Eur J Sport Sci*. 2014;14:772–781. PubMed doi:10.1080/17461391.2014.905987
4. Cormie P, McGuigan MR, Newton RU. Developing maximal neuromuscular power: part 2 - training considerations for improving maximal power production. *Sports Med*. 2011;41:125–146. PubMed doi:10.2165/11538500-000000000-00000
5. Haff GG, Hobbs RT, Haff EE, Sands WA, Pierce KC, Stone MH. Cluster training: a novel method for introducing training program variation. *Strength Condit J*. 2008;30:67–76. doi:10.1519/SSC.0b013e31816383e1
6. Haff GG, Burgess S, Stone M. Cluster training: theoretical and practical applications for the strength and conditioning professional. *Prof Strength Cond.* 2008;12:12–17.
7. Haff GG, Whitley A, McCoy LB, et al. Effects of different set configurations on barbell velocity and displacement during a clean pull. *J Strength Cond Res*. 2003;17:95–103. PubMed
8. Gorostiaga EM, Navarro-Amezqueta I, Calbet JA, et al. Blood ammonia and lactate as markers of muscle metabolites during leg press exercise. *J Strength Cond Res*. 2014;28:2775–2785. PubMed doi:10.1519/JSC.0000000000000496
9. Gorostiaga EM, Navarro-Amezqueta I, Calbet JA, et al. Energy metabolism during repeated sets of leg press exercise leading to failure or not [published correction appears in *PLoS One*. 2013;8(1): doi:10.1371/annotation/ca21efee-84a1-4031-9dcd-224af64b7753]. *PLoS One*. 2012;7:e40621. PubMed doi:10.1371/journal.pone.0040621
10. Hardee JP, Triplett NT, Utter AC, Zwetsloot KA, McBride JM. Effect of interrepetition rest on power output in the power clean. *J Strength Cond Res*. 2012;26:883–889. PubMed doi:10.1519/JSC.0b013e3182474370
11. Iglesias-Soler E, Carballeira E, Sanchez-Otero T, Mayo X, Jimenez A, Chapman ML. Acute effects of distribution of rest between repetitions. *Int J Sports Med*. 2012;33:351–358. PubMed doi:10.1055/s-0031-1299699
12. Moreno SD, Brown LE, Coburn JW, Judelson DA. Effect of cluster sets on plyometric jump power. *J Strength Cond Res*. 2014;28:2424–2428. PubMed doi:10.1519/JSC.0000000000000585
13. Gorostiaga EM, Navarro-Amezqueta I, Cusso R, et al. Anaerobic energy expenditure and mechanical efficiency during exhaustive leg press exercise. *PLoS One*. 2010;5:e13486. PubMed doi:10.1371/journal.pone.0013486
14. Sánchez-Medina L, Gonzalez-Badillo JJ. Velocity loss as an indicator of neuromuscular fatigue during resistance training. *Med Sci Sports Exerc*. 2011;43:1725–1734. PubMed doi:10.1249/MSS.0b013e318213f880
15. Hansen KT, Cronin JB, Pickering SL, Newton MJ. Does cluster loading enhance lower body power development in preseason preparation of elite rugby union players? *J Strength Cond Res*. 2011;25:2118–2126. PubMed doi:10.1519/JSC.0b013e318220b6a3
16. Oliver JM, Kreutzer A, Jenke SC, Phillips MD, Mitchell JB, Jones MT. Velocity drives greater power observed during back squat using cluster sets. *J Strength Cond Res.* 2016;30(1):235–243.
17. Iglesias-Soler E, Carballeira E, Sanchez-Otero T, Mayo X, Fernandez-Del-Olmo M. Performance of maximum number of repetitions with cluster-set configuration. *Int J Sports Physiol Perform*. 2014;9:637–642. PubMed doi:10.1123/ijspp.2013-0246
18. Hardee JP, Lawrence MM, Utter AC, Triplett NT, Zwetsloot KA, McBride JM. Effect of inter-repetition rest on ratings of perceived exertion during multiple sets of the power clean. *Eur J Appl Physiol*. 2012;112:3141–3147. PubMed doi:10.1007/s00421-011-2300-x
19. Denton J, Cronin JB. Kinematic, kinetic, and blood lactate profiles of continuous and intraset rest loading schemes. *J Strength Cond Res*. 2006;20:528–534. PubMed
20. Joy JM, Oliver JM, McCleary SA, Lowery RP, Wilson JM. Power output and electromyography activity of the back squat exercise with cluster sets. *J Sports Sci*. 2013;1:37–45.
21. Lawton TW, Cronin JB, Lindsell RP. Effect of interrepetition rest intervals on weight training repetition power output. *J Strength Cond Res*. 2006;20:172–176. PubMed
22. Moir GL, Graham BW, Davis SE, Guers JJ, Witmer CA. Effect of cluster set configurations on mechanical variables during the deadlift exercise. *J Hum Kinet*. 2013;39:15–23. PubMed
23. Oliver JM, Kreutzer A, Jenke S, Phillips MD, Mitchell JB, Jones MT. Acute response to cluster sets in trained and untrained men. *Eur J Appl Physiol*. 2015;115:2383–2393. PubMed doi:10.1007/s00421-015-3216-7
24. Matuszak ME, Fry AC, Weiss LW, Ireland TR, McKnight MM. Effect of rest interval length on repeated 1 repetition maximum back squats. *J Strength Cond Res*. 2003;17:634–637. PubMed
25. Cormie P, McCaulley GO, Triplett NT, McBride JM. Optimal loading for maximal power output during lower-body resistance exercises. *Med Sci Sports Exerc*. 2007;39:340–349. PubMed doi:10.1249/01.mss.0000246993.71599.bf
26. Cormie P, McBride JM, McCaulley GO. Validation of power measurement techniques in dynamic lower body resistance exercises. *J Appl Biomech*. 2007;23:103–118. PubMed doi:10.1123/jab.23.2.103
27. Folland JP, Irish CS, Roberts JC, Tarr JE, Jones DA. Fatigue is not a necessary stimulus for strength gains during resistance training. *Br J Sports Med*. 2002;36:370–373. PubMed doi:10.1136/bjsm.36.5.370
28. Drinkwater EJ, Lawton TW, McKenna MJ, Lindsell RP, Hunt PH, Pyne DB. Increased number of forced repetitions does not enhance strength development with resistance training. *J Strength Cond Res*. 2007;21:841–847. PubMed
29. Girman JC, Jones MT, Matthews TD, Wood RJ. Acute effects of a cluster-set protocol on hormonal, metabolic and performance measures in resistance-trained males. *Eur J Sport Sci*. 2014;14:151–159. PubMed doi:10.1080/17461391.2013.775351
30. Mayo X, Iglesias-Soler E, Fernández-Del-Olmo M. Effects of set configuration of resistance exercise on perceived exertion. *Percept Mot Skills*. 2014;119:825–837. PubMed doi:10.2466/25.29.PMS.119c30z3